Supplement S2: Pipeline documentation

Registered Report: Is Algorithm Aversion a Rule or a Weight?

This supplement documents the technical components and specifications of the implemented pilot. The questionnaire, the interviewer prompt and the coding protocol are archived on GitHub (`https://github.com/dkarpa/lata-registered-report`) and archived on Zenodo under a CC0 license at doi:10.5281/zenodo.23362235; the stimulus code and the Qualtrics file follow with the Stage 2 submission.

1 Survey platform

- Platform: Mobile-only Qualtrics survey.
- Languages: Finnish, Swedish, English.
- Survey Flow: A server-side Web Service calls `POST /v1/session/start` with the study, survey and Qualtrics session IDs and a study secret; its returned upload token is mapped to Embedded Data before recorded pages load. The current OASIS pilot embeds an interview in Part 6.
- Embedded Data fields: `SurveyID` identifies the survey; `SessionID` groups and links trace files to the Qualtrics response and is passed to OASIS as `pid`; `ResponseID` is the Qualtrics response identifier. `OASIS_Status`, `OASIS_EndReason` and `OASIS_EndTimeUTC` describe the interview outcome.

2 Conjoint stimulus

- Design: six tasks, two systems per task, five attributes with three levels; levels drawn independently and uniformly, with one restriction: if the second system equals the first on all five attributes, it is redrawn (`genDesign`), so identical pairs never occur; attribute row order randomized across respondents and fixed within respondent.
- Reveal: cells are hidden until tapped (a "+" is shown) and then stay open. Each opening is logged with a millisecond timestamp.
- Responses: binary choice (System A/B) and the compliance slider (1–7).
- Payload (Saved Conjoint Response): One JSON object per respondent, built in the essay field after all 6 tasks are completed; truncation rule if the payload exceeds 19,000 characters.

3 Interaction recorder

- What is recorded: rrweb DOM snapshots and changes, pointer movement (sampled every 100 ms), scrolling (150 ms), taps and interaction events (such as selection state).
- Batching: Chunks are uploaded every 4 seconds; full snapshot every 60 seconds. Page submit or unload attempts a final chunk. Not recorded: anything inside the embedded interview.
- Consent gate: The client starts only when consent is obtained, i.e., from Page 2 of the survey, after the consent form.
- Storage and retention: gzip JSON chunks on the University of Helsinki's secure servers. The folders are grouped by study, `SessionID` and segment. Separate .py files to inspect chunk

order, gaps, snapshots and final markers. In addition another .py file derives replay JSON, and the semantic events are exported to a CSV.
- Retention: as stated in the data protection notice, personal data are processed until the end of the project (31 December 2027); the data are then anonymized, remaining identifiable data are destroyed, and the anonymized data, including the interaction logs, are deposited at the Finnish Social Science Data Archive (FSD).
- Linkage: records are linked to the Qualtrics `SessionID`.

4 LLM interviewer

- Platform: OASIS text interview, free-form mode, self-hosted. The pilot ran on a local Docker Compose deployment (Caddy, backend, PostgreSQL, Redis and Ollama); in the field it will run on a CSC GPU server with the same model and configuration.
- Model: local Ollama tag `qwen3:4b-instruct-2507-q4_k_m`. The temperature is set to 0.7 and the maximum duration of the interview will be 10 minutes.
- Prompt: identical English system prompt in all three survey versions (Supplement S1); OASIS language setting per version (Finnish, Swedish, English); fixed welcome message, final questions and closing sentence in each language.
- Logging: OASIS stores session metadata (including participant ID, status and duration) in sessions, with ordered role, content, timestamp and available token counts in `transcript_entries`.

5 Transcript coder

- Model choices: `Qwen3-4B`, `Qwen3-30B-A3B`, `Llama-4-Scout-17B-16E-Instruct` and `Llama-3.3-70B`. We consider four open-weight models that Fuławka et al. (2026) also tested, while also ensuring that the models can be hosted on CSC and provide open weights for secure and local processing. The Qwen3 models are described in Yang et al. (2025), and Llama-3.3-70B belongs to the Llama 3 family (Grattafiori et al., 2024). Step 1 of the analysis plan chooses among them; a tie goes to the smaller model.

6 Additive preference model (H2)

- R package `sconjoint` (Acharya et al., 2026), most recent release at the time of analysis (version reported at Stage 2), with its default settings; cross-fitting by respondent and a leave-one-task-out update as specified in the main text.
- Covariates: all items of Parts 1–4 (Supplement S1): mood, positive and negative attitudes toward AI, appropriateness of AI, support for welfare reforms, generalized and institutional trust, risk attitude, vote intention, party sympathy, left–right placement, birth year, gender, education, place of residence, household income and home language; categorical items enter as indicators, with "Don't know" as its own category.

7 Panel quality controls

Norstat recruits panel members by invitation only and verifies their email addresses and phone numbers, to keep out bots and duplicate accounts. It monitors responses for careless or inattentive answering, checks suspicious responses manually, and removes members who do not meet its quality standards (`https://norstat.co/how-it-works/consumer-panels/quality/`, accessed 14 October 2026).

8 Reproducibility package

- Materials repository: on GitHub (`https://github.com/dkarpa/lata-registered-report`) and archived on Zenodo under a CC0 license at doi:10.5281/zenodo.23362235.

References

Acharya, A., Hainmueller, J., and Xu, Y. (2026). Learning preferences from conjoint data: A hybrid structural deep learning approach. arXiv:2604.10845. Working paper, Stanford University, 29 June 2026.

Fuławka, K., Hertwig, R., and Wulff, D. U. (2026). Large language models accurately identify decision reasons in verbal reports. *Proceedings of the National Academy of Sciences*, 123(27):e2526798123.

Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., et al. (2024). The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

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